SERV 1.4.0
Development study

PAUSED RESEARCH | 12 DEVELOPMENT PAIRS | FINAL ACCEPTANCE PENDING

Record prepared 13 October 2026. The study was paused by the user on 12 October 2026. This package preserves the best saved implementation and its evidence. It is a research record alongside the three accepted portfolio cases.

Best saved result

Metric	Baseline mean	Optimized mean	Paired change
Total area	2,743,003.5 MWTA	2,579,889.0 MWTA	5.9467% lower
Critical path	4.275710 ns	4.278436 ns	0.0721% higher
Active total power	3.541333 mW	3.386167 mW	4.3802% lower
Packed CLBs	46	43	6.5217% fewer
Mapped flip-flops	181	181	Unchanged
Composite PPA	1.000000	0.965484	3.4516% lower

Metric means are arithmetic means. Paired changes use geometric means of per-pair ratios. The composite equally weights the log area, delay and power ratios. A small timing increase remains visible.

Scope of the result

The estimates use the pinned academic VTR/PTM 45 nm flow, a homogeneous LUT6 target and ACE probabilistic activity at 0.9 V and 85 C. MWTA is the minimum-width transistor-area unit, not square millimetres. The result is not measured silicon, board power, workload energy, ASIC signoff or commercial-FPGA characterization.

3.4516% is a development selection result. The current candidate has no held-out certification result and no final 64-pair result. The original greater-than-10% objective remains unmet.

Design and implementation

Frozen measurement boundary

SERV is a bit-serial 32-bit RISC-V processor. The top is serv_synth_wrapper with PRE_REGISTER=1, RESET_STRATEGY=MINI, WITH_CSR=1, RF_WIDTH=2, W=1, MDU=0, COMPRESSED=0 and ALIGN=0. Instruction/data buses and register-file addresses, data and handshakes are part of the protected boundary.

The baseline contains eighteen files, all byte-identical to the pinned SERV 1.4.0 commit. The optimized tree also has eighteen files; the exact patch changes ten. Total source length is 3,562 baseline lines and 4,137 optimized lines, including comments, blank lines and inactive parameter alternatives. Source length is not a physical area estimate.

What the combined candidate changes

The candidate combines a three-bit representation of the high register-file phase with earlier decoder, buffer, PC and control rewrites. The frozen wrapper selects RF_HIGH_PHASE_FORM=1, DECODER_CAPTURE_USAGE_FORM=3, PC_WORD_ENABLE_FORM=3, DATA_WORD_ENABLE_FORM=3 and BUFFER_CONTROL_FORM=3. The result belongs to this complete selected implementation; component percentages cannot be added or attributed independently.

Capacity and physical state

Quantity	Baseline	Optimized
GPR words / RAM-backed CSR words	32 / 4	32 / 4
Bits per register word	32	32
External register storage	1,152 bits / 144 bytes	1,152 bits / 144 bytes
Logical rows / SRAM interface	576 rows / 2 bits	576 rows / 2 bits
Mapped wrapper state	181 flip-flops	181 flip-flops

The external SRAM is excluded from mapped wrapper area. Its storage capacity is unchanged. The wrapper flip-flop count and external register-file storage describe different resources.

Correctness records

Saved source gates include lint at W=1 and W=4 and sequential equivalence with 357 proved / zero unproved obligations. RTL-to-measured-netlist equivalence covers 126 primary output bits and all 181 state bits: 307 proved / zero unproved for each design. Undefined initialization is preserved in the proof.

Two recorded 4,096-cycle simulations per design compare defined RTL output bits; unknown RTL bits are skipped. These simulations supplement the formal record. The evidence establishes RTL-boundary equivalence and does not certify full RISC-V ISA compliance.

Search and stopping point

Closed revision-2 exploration

Recorded activity	Count / status
Registered exploratory candidates	2,812
Candidates with valid PPA measurements	2,621
Candidates excluded without PPA	191
Selected RF-phase cohort	36 measured candidates
Best candidate paired development comparison	12 pairs / 24 implementations
Held-out certification / final publication	0 / 0 completed pairs for current leader

Exploration counts are not independent acceptance trials. Later searches used the best candidate and other promising parents, including combinations on intermediate STATE variants. No completed validated combination superseded this leader.

Flow repair and historical results

The adopted revision-2 flow lowers subtraction through standard Yosys mapping before Parmys, applied equally to baseline and candidates. This repair followed a discrepancy between source RTL and a measured mapped implementation. Historical measurements are kept separate from the present result.

An apparent 11.10461% gain was rejected for mapping incoherence. The historical 4.4778% gain was not recertified after the flow repair. Two older exploratory 64-pair replicas are not final certification of this candidate. None is promoted as the current publication result.

Last incomplete family

Nine branch/trap variants were prepared. Thirty-six full elaborations and three native banks completed, with 786,624 configuration-clocks checked. Six native banks remained pending. No PPA measurement was completed for that family; no improvement is assigned to it.

Original acceptance rule remains pending

The study reserved a held-out twelve-pair certification sample and a final 64-pair publication sample. Both remain unexecuted for the leader. The original editorial rule requires a point score and one-sided 95% upper score bound strictly below 0.90, plus the declared primary-metric and multiplicity rules. A descriptive interval from selected development data is not a confirmatory test.

The user paused the search after repeated storage stops. The best source and evidence are preserved. Registering this paused research record does not change the experimental acceptance rule or turn it into an accepted fourth portfolio case.

Evidence and verification

Public package

The package includes baseline and optimized RTL, an exact unified patch, a contract projection, upstream provenance, the original leader and pause records, projected measurement records, all 24 timing/area and power reports, flow logs and measured BLIF netlists. It also includes saved source lint/formal logs, mapped formal logs/models/scripts and simulation completion logs.

The original development audit is preserved. The full exploration archive, optimizer and private execution infrastructure are outside this package. No downloadable full exploration archive or newly executed EDA replay is claimed.

Private host path prefixes in four simulation logs are replaced by a stable label. Their projection provenance records original and public hashes; all other bytes are preserved.

Run the saved-evidence verifier

```
python3 experiments/serv-1.4.0/verify.py
python3 verify.py
```

The verifier checks exact source identities, upstream Git blob identities, the patch, package checksums, fixed evidence authorities, proof closure markers, measured netlist identities and the arithmetic recomputed from the saved raw reports. It also checks that certification and final-pair completion remain false.

Expected result: saved research evidence PASS; development_pairs=12; improvement=3.4516%; held_out_certification_pairs=0; final_pairs=0; publication_accepted=False.

Verification scope: consistency of saved evidence. It does not rerun EDA, independently regenerate a formal proof, certify PPA, or constitute external assurance.

Pinned identities

```
SERV upstream commit
7d9cde4e6ca4f4c84d7512a752a4c187537dd373

Baseline RTL tree SHA-256
29b5ea77526b3b972ba99c8702524e87035e5a921db6f1a59bb822599203ba91

Optimized RTL tree SHA-256
b25b924a34318591d7cfec7097fc5d265a372316757893b2d51a095c5ba01fbe

VTR commit
95f5c6de9e158371ba7185bf97c07a84153735d6
```

Source, licensing and further reading

Baseline and derived RTL retain upstream ISC headers. The upstream ISC license is bundled. Original presentation and verification work follows the repository MIT license. SERV, its authors and tool providers do not endorse this study.

Primary source: github.com/olofk/serv at the commit above. Package references: README.md, contract.json, provenance.json, results/development.json, results/measurements.json and results/pause-record.json. Repository methodology and audit scope: METHODOLOGY.md and AUDIT.md.